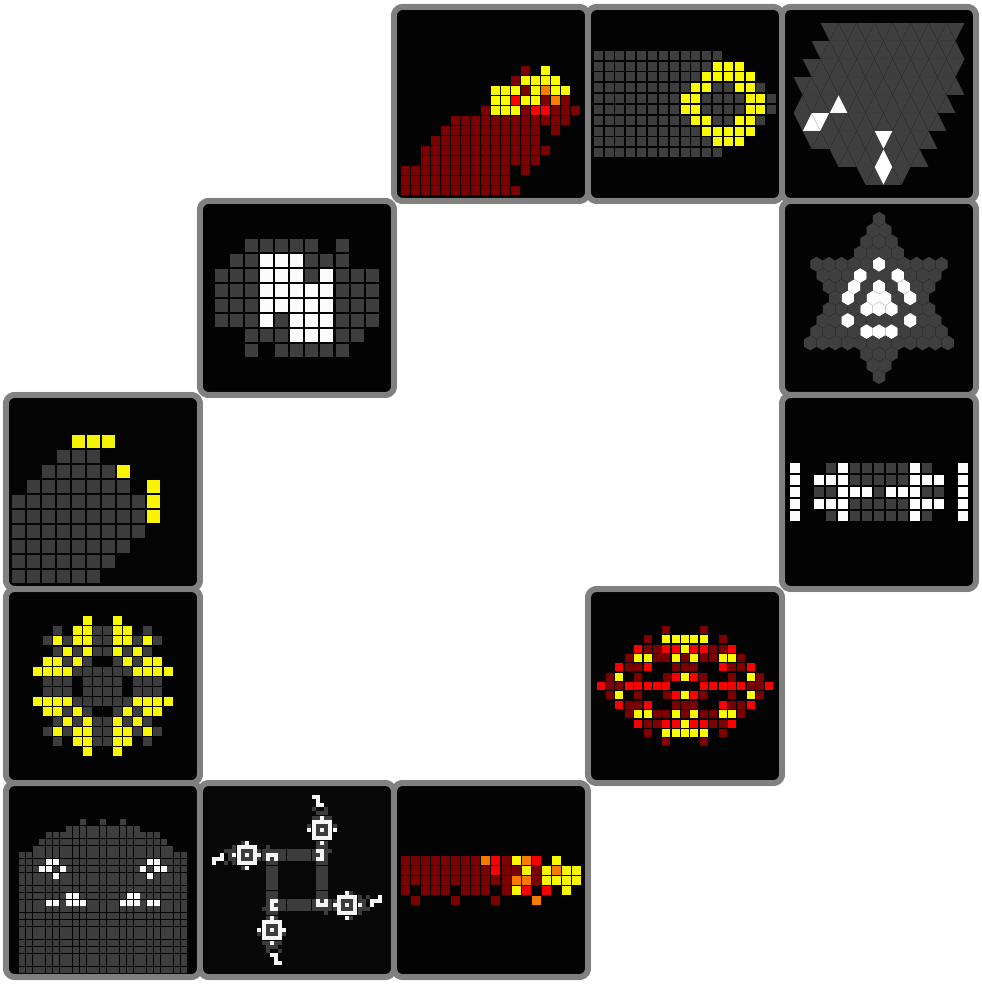
A World of Cellular Automata

Beneath the Surface

iniquity816



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April Fools!!!!!!!!1

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Preface

Intentions

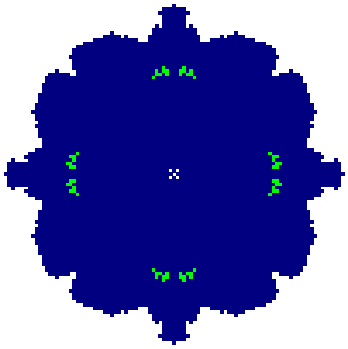
This book is intended to be an unofficial counterpart to *Conway's Game of Life: Mathematics and Construction* by Nathaniel Johnston and Dave Greene (hereinafter "the Life book"), focusing on the large variety of cellular automata that allow for elementary objects and mechanisms that Life simply does not have.[1] Rules with a square tiling, being the most ubiquitous manifestation of cellular automata rules, are well-explored and are thus the main object of observation of this book, but rules on other, more novel tilings (such as the hexagonal and triangular tilings) are also included to present the world of cellular automata more objectively, and to perhaps foster exploration and a better understanding of these rules. Rules with larger ranges, such as those explored by Kellie Evans, are also present, but note that these rules become less easier to simulate as the range of the neighborhood, or the area of cells that can affect one cell, increases. The maximum range that can be simulated in most programs is 500.

Prerequisites

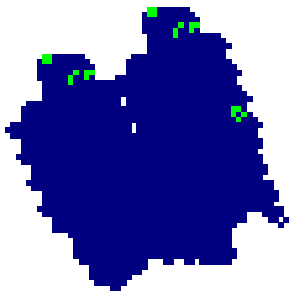
It is assumed that you have the program Golly installed, available here, which natively facilitates the simulation of almost all rulespaces discussed in this book and enables the creation of scripts to achieve certain goals. Additionally, it is assumed that you are familiar with the Life book, which discusses ideas that mostly overlap with this book. Nothing else is required to use each program comfortably, but web resources can be used to broaden your knowledge if necessary.

Usage

Much like the Life book, this book is reliant on figures to illustrate patterns in a non-changing manner, which makes the visualisation of spaceships and other moving objects more difficult. Hence, cells that were alive in previous generations but are now dead (the "envelope" of the pattern) are highlighted blue to indicate change. This ensures the appearance of motion in moving objects and volatility in oscillators. Still lives, showing no movement or change from one generation to the next, are presented "as-is", without highlighting of any kind.



(a) Period-661 oscillator in `B356/S23`.



(b) (5,2)*c*/190 spaceship in `B38/S23`, run for 760 generations.

Examples of envelopes for an oscillator and a spaceship (each in their minimal phase, in terms of population), coloured using LifeHistory state colours.

Cells may also be given different colours as part of an exercise, which we will discuss below.

[1] Any object in any rule with a range-1 Moore neighbourhood, like Conway's Game of Life, can be emulated in Life using 0E0P metacells.

Exercises

Exercises are also used to test one's knowledge and are placed at the end of each chapter. They are assigned difficulty ratings, ranging from Difficulty-1 (straightforward and often completed without much effort) to Difficulty-5 (projects that may take a day or two to complete). It is hoped that these exercises improve the breadth of one's knowledge and experience.

Acknowledgements

First and foremost, I would like to thank Nathaniel Johnston and Dave Greene, who directly inspired this book and indirectly motivated me to write it. I extend thanks to David S. Miller, the forum user "FWKnightship", and the creators of Golly (Andrew Trevorrow and Tomas Rokicki). Finally, I would like to thank Mathias Legrand and Velimir Gayevskiy, creators of the template *Legrand Orange Book*, available at LaTeXTemplates.com.

The Square Tiling

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1. The von Neumann Neighbourhood

The von Neumann neighbourhood, named after none other than John von Neumann (1903–1957), limits the number of neighbours a single cell can have to just the cells orthogonally adjacent to it. A 29-state rule with the von Neumann neighbourhood, devised by von Neumann in the 1940s with suggestions from Stanisław Ulam (1909–1984) to realise a **universal constructor** in the form of a **self-replicating machine**, was the first example of a cellular automaton using this neighbourhood (and the first cellular automaton in general). The first implementation of his machine, using his original 29-state specification, would only come in 2007.[1]

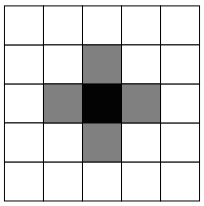

Figure 1.1: The von Neumann neighbourhood (in grey) of the central cell (in **black**).

1.1 Two-state von Neumann Rules

Two-state von Neumann rules are extremely limited; they require birth on 1 neighbour to expand outside their initial bounding box (but this, in turn, makes rules hopelessly explosive). A simple proof of this is provided below. [add proof]

1.2 Three-state von Neumann Rules

[insert shameless self plug here]

1.3 Circuitry Rules

All four Banks rules, Codd, Devore, Flow6, and von Neumann's constructor rule all use the von Neumann neighbourhood. [add pictures of Banks-I to Banks-IV, Codd, Devore, Flow6, and the Sphinx in action]

[1] An earlier implementation by Renato Nobili and Umberto Pesavento in 1994 used a version of von Neumann's rule with 32 states. [**10.1162/artl.1995.2.4.337**]

2. The Moore Neighbourhood

2.1 Outer-Totalistic Rules

In outer-totalistic rules, only the number of alive cells a cell has matters (cells are born without consideration of the positions of the neighbours of the cell, relative to eachother). The Game of Life belongs to this family, and so does the variants Grounded Life (`B35/S23`), HighLife (`B36/S23`), DryLife (`B37/S23`), Pedestrian Life (`B38/S23`), EightLife (`B3/S238`), and HoneyLife (`B38/S238`). 2^{18} rules are in this family.

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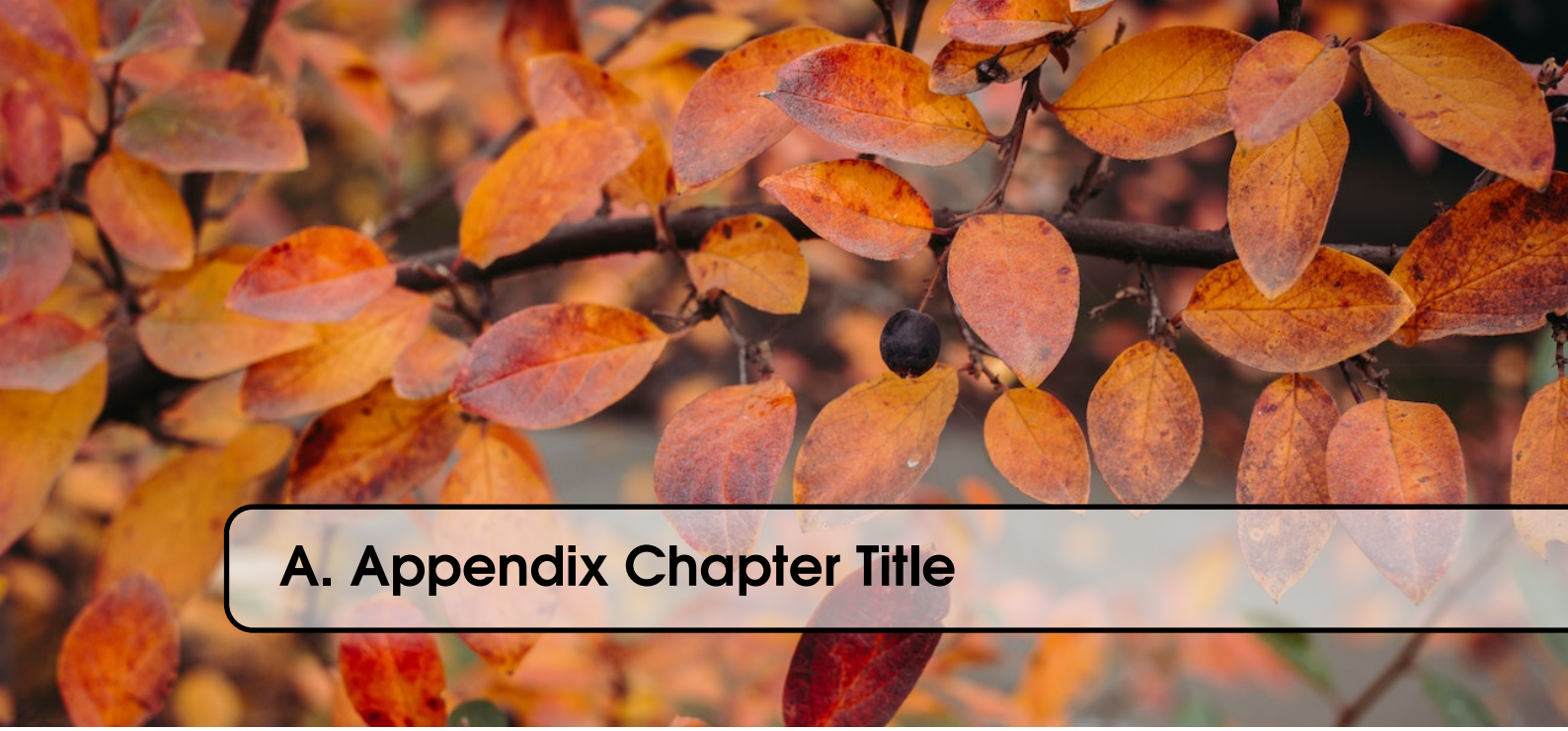

A. Appendix Chapter Title

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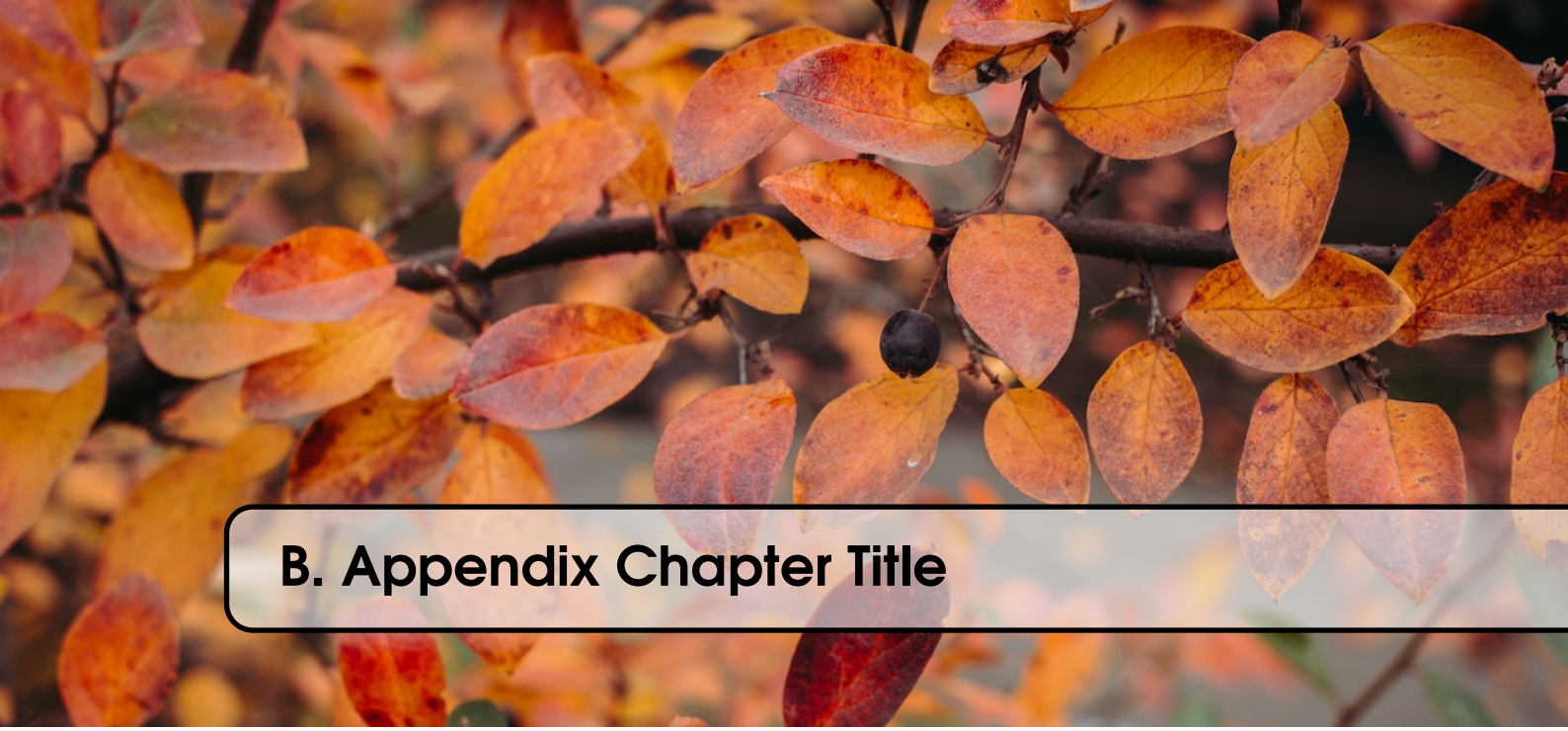

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